

DERIVATIVES PRICING

A COMPUTATIONAL & THEORETICAL FRAMEWORK

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Part I

Mathematical Foundations

1 Measure and Integration

2 Probability

2.1 Modelling Random Experiments

Probability theory is about modelling and analysing *random experiments*: experiments whose outcome cannot be determined in advance, but are nevertheless still subject to analysis. Mathematically, we can model a random experiment by defining a specific measure space $(\Omega, \mathcal{H}, \mathbb{P})$, which we call a *probability space*, where the three components are: a *sample space*, a collection of *events*, and a *probability measure*.

The *sample space* Ω of a random experiment is the set of all possible outcomes of the random experiment. Sample spaces can be countable, such as $\mathbb{N}$, or uncountable, such as $\mathbb{R}^n$ or $\{0, 1\}^\infty$. The sets in the σ -algebra $\mathcal{H}$ are called *events*. These are the subsets to which we wish to assign a probability. We can view $\mathcal{H}$ as the family of events. An event A *occurs* if the outcome of the experiment is one of the elements in A . The properties of $\mathcal{H}$ are natural in the context of random experiments and events:

- If A and B are events, the set $A \cup B$ is also an event, namely the event that A *or* B or both occur. For a sequence $A_1, A_2, \dots$ of events, their union is the event that *at least one* of the events occurs.
- If A and B are events, the set $A \cap B$ is also an event, namely the event that A *and* B both occur. For a sequence $A_1, A_2, \dots$ of events, their intersection is the event that *all* of the events occur.
- If A is an event, its complement A^c is also an event, namely the event that A *does not* occur.

- The set Ω itself is an event, namely the *certain event* (it always occurs). Similarly, $\emptyset$ is an event, namely the *impossible event* (it never occurs).

When the sample space Ω is $\mathbb{R}$, the natural σ -algebra $\mathcal{H}$ is the *Borel σ -algebra* $\mathcal{B}$. Recall that this is the smallest σ -algebra on $\mathbb{R}$ that contains all the intervals of the form $(-\infty, x]$ for $x \in \mathbb{R}$. This σ -algebra of sets is big enough to contain all important sets and small enough to allow us to assign a probability to all events.

The third ingredient in the model for a random experiment is the specification of the probability of the events. In fact, this is the crucial part of the model. Mathematically, we are looking for a *measure* $\mathbb{P}$ that assigns to each event A a number $\mathbb{P}(A)$ in $[0, 1]$ describing how likely or probable that event is.

Definition 2.1: Probability Measure

A *probability measure* is a measure on $(\Omega, \mathcal{H})$ with $\mathbb{P}(\Omega) = 1$. Thus, $\mathbb{P}$ is a mapping from $\mathcal{H}$ to $[0, 1]$ with the following properties:

- (1) $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$.
- (2) For any sequence $A_1, A_2, \dots$ of disjoint events,

$$\mathbb{P}(\cup_n A_n) = \sum_n \mathbb{P}(A_n). \quad (2.1)$$

An event A is said to occur *almost surely* (a.s.) if $\mathbb{P}(A) = 1$.

Theorem 2.1: Properties of a Probability Measure

Let $\mathbb{P}$ be a probability measure on a measurable space $(\Omega, \mathcal{H})$, and let A, B and $A_1, A_2, \dots$ be events. Then, the following hold:

- (1) (*Monotonicity*): $A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$.
- (2) (*Continuity from below*): If $A_1, A_2, \dots$ is an increasing sequence of events, i.e., $A_1 \subseteq A_2 \subseteq \dots$, then

$$\lim \mathbb{P}(A_n) = \mathbb{P}(\cup_n A_n). \quad (2.2)$$

- (3) (*Countable subadditivity*): $\mathbb{P}(\cup_n A_n) \leq \sum_n \mathbb{P}(A_n)$.

Theorem 2.2: Additional Properties of a Probability Measure

Let $\mathbb{P}$ be a probability measure on a measurable space $(\Omega, \mathcal{H})$, and let A, B and $A_1, A_2, \dots$ be events. Then, the following hold:

- (1) (Complement): $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- (2) (Union): $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- (3) (Continuity from above): If $A_1, A_2, \dots$ is a decreasing sequence of events, i.e., $A_1 \supseteq A_2 \supseteq \dots$, then

$$\lim \mathbb{P}(A_n) = \mathbb{P}(\cap_n A_n). \quad (2.3)$$

2.2 Random Variables

Usually, the most convenient way to describe quantities of interest connected with random experiments is by using random variables. Random variables allow us to use intuitive notations for certain events, such as $\{X \in A\}$, etc. Intuitively, we can think of a random variable X as a measurement on a random experiment that will become available *tomorrow*. However, all the think work can be done *today*. If for each possible outcome ω , the measurement is $X(\omega)$, we can specify today probabilities such as $\mathbb{P}(X \in A)$.

We can translate our intuitive notion of a random variable into rigorous mathematics by defining a random variable to be a measurable function from $(\Omega, \mathcal{H})$ to a measurable space $(E, \mathcal{E})$.

Definition 2.2: Random Variable

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and $(E, \mathcal{E})$ a measurable space. A *random variable* X taking values in E is an $\mathcal{H}/\mathcal{E}$ -measurable function; that is, a mapping $X : \Omega \rightarrow E$ that satisfies

$$X^{-1}A = \{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{H} \quad \text{for all } A \in \mathcal{E}. \quad (2.4)$$

Often a random experiment is described via more than one random variable. Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and let $\mathbb{T}$ be an arbitrary index set, which may be countable or uncountable. Think of $\mathbb{T}$ as a set of times. For each $t \in \mathbb{T}$, let $(E_t, \mathcal{E}_t)$ be a measurable

space. The product measurable space of $\{(E_t, \mathcal{E}_t) : t \in \mathbb{T}\}$ is

$$\left(\times_{t \in \mathbb{T}} E_t, \bigotimes_{t \in \mathbb{T}} \mathcal{E}_t \right).$$

Each element $x \in \times_{t \in \mathbb{T}} E_t$ is of the form $x = (x_t)_{t \in \mathbb{T}}$. Often $(E_t, \mathcal{E}_t)$ is the same space $(E, \mathcal{E})$ for all t , in which case x can be viewed as a function $x : \mathbb{T} \rightarrow E$. The σ -algebra on $\times_{t \in \mathbb{T}} E_t$ is generated by the rectangles

$$R((A_t)_{t \in \mathbb{T}}) := \left\{ x \in \times_{t \in \mathbb{T}} E_t : x_t \in A_t \text{ for each } t \in \mathbb{T} \right\}, \quad (2.5)$$

where $A_t \in \mathcal{E}_t$ and $A_t = E_t$ except for finitely many t .

Definition 2.3: Stochastic Process

A collection of random variables $X := \{X_t, t \in \mathbb{T}\}$, where $\mathbb{T}$ is any index set, is called a *stochastic process*. When $\mathbb{T}$ is finite, X is called a *random vector*.

Theorem 2.3: Stochastic Process

$X := \{X_t, t \in \mathbb{T}\}$ is a stochastic process taking values in $\times_{t \in \mathbb{T}} E_t$ with σ -algebra $\bigotimes_{t \in \mathbb{T}} \mathcal{E}_t$ if and only if each X_t is a random variable taking values in E_t with σ -algebra $\mathcal{E}_t$.

2.3 Probability Distributions

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and let X be a random variable taking values in a set E with σ -algebra $\mathcal{E}$. To simplify our usage, we also say that “ X takes values in $(E, \mathcal{E})$ ”. If we wish to describe our experiment via X , then we need to specify the probabilities of events such as $\{X \in B\}$, $B \in \mathcal{E}$. In elementary probability theory we usually made a distinction between “discrete” and “continuous” random variables. However, it is sometimes better to analyse random variables in a more unified manner; many definitions and properties of random variables do not depend on whether they are “discrete” or “continuous”. Moreover, some random variables are neither discrete nor continuous.

Recall that by definition *all* the sets $\{X \in B\}$ with $B \in \mathcal{E}$ are events. So we may assign a probability to each of these events. This leads to the notion of the *distribution* of a random variable.

Definition 2.4: Probability Distribution

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and let X be a random variable taking values in $(E, \mathcal{E})$. The (*probability*) *distribution* of X is the image measure μ of $\mathbb{P}$ under X . That is,

$$\mu(B) := \mathbb{P}(X \in B), \quad B \in \mathcal{E}. \quad (2.6)$$

Definition 2.5: Cumulative Distribution Function

The (*cumulative*) *distribution function* of a numerical random variable X is the function $F : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F(x) := \mathbb{P}(X \leq x), \quad x \in \mathbb{R}. \quad (2.7)$$

Proposition 2.1: Properties of a Cumulative Distribution Function

- (1) (*Bounded*): $0 \leq F(x) \leq 1$.
- (2) (*Increasing*): $x \leq y \implies F(x) \leq F(y)$.
- (3) (*Right-continuous*): $\lim_{h \downarrow 0} F(x + h) = F(x)$.

Definition 2.6: Discrete Distribution

We say that X has a *discrete distribution* μ on $(E, \mathcal{E})$ if μ is a discrete measure; that is, it is of the form

$$\mu = \sum_{x \in D} f(x) \delta_x$$

for some countable set D and positive masses $\{f(x), x \in D\}$.

Definition 2.7: Distribution with a Density

We say that X has an *absolutely continuous* distribution μ on $(E, \mathcal{E})$ with respect to a measure λ on $(E, \mathcal{E})$ if there exists a positive $\mathcal{E}$ -measurable function f such that $\mu = f\lambda$; that is,

$$\mu(B) = \mathbb{P}(X \in B) = \int_B \lambda(dx) f(x), \quad B \in \mathcal{E}.$$

The function f is thus the (*probability*) *density function* (pdf) of X with respect to the measure, λ often taken to be the Lebesgue measure. Note that μ is absolutely continuous with respect to λ in the sense that for all $B \in \mathcal{E} : \lambda(B) = 0 \implies \mu(B) = 0$.

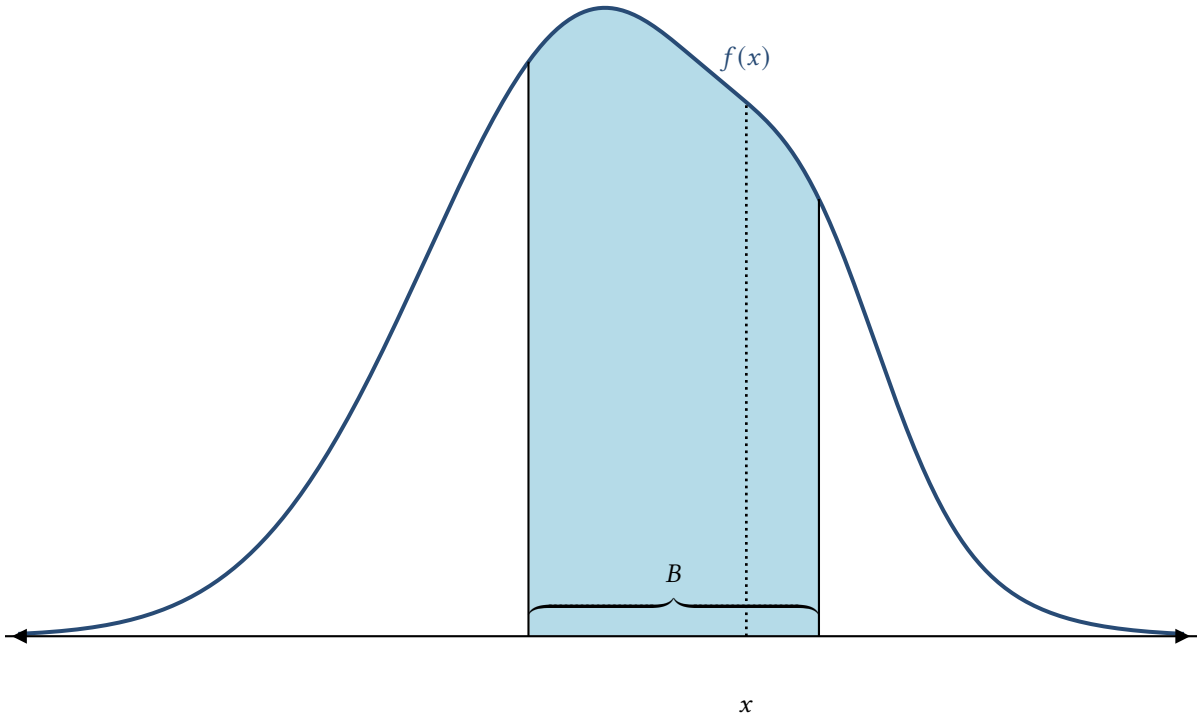


Figure 1: A probability density function of a numerical random variable

Describing an experiment via a random variable and its probability distribution or density is often more convenient than specifying the probability space. In fact, the probability space usually stays in the background. The question remains, however, whether there *exists* a probability space $\Omega, \mathcal{H}, \mathbb{P}$ and a numerical random variable X for a given probability distribution. Fortunately, existence can be established for all practical probability models.

The concept of independence is of great importance in the study of random experiments and the construction of probability distributions. Loosely speaking, independence is about the lack of shared information between random objects.

Definition 2.8: Independent Random Variables

Let X and Y be random variables taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$, respectively. They are said to be *independent* if for any $A \in \mathcal{E}$ and $B \in \mathcal{F}$ it holds that

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B). \quad (2.8)$$

Intuitively, this means that information regarding X does not affect our knowledge of Y and vice versa. Mathematically, Equation (2.8) simply states that the distribution, μ say, of (X, Y) is given by the *product measure* of the distributions μ_X of X and μ_Y of Y . We can extend the independence concept to a finite or infinite collection of random variables as follows:

Definition 2.9: Independency

We say that a collection of random variables $\{X_t, t \in \mathbb{T}\}$, with each X_t taking values in $(E_t, \mathcal{E}_t)$, is an *independency* if for any finite choice of indexes $t_1, \dots, t_n \in \mathbb{T}$ and sets $A_{t_1} \in \mathcal{E}_{t_1}, \dots, A_{t_n} \in \mathcal{E}_{t_n}$ it holds that

$$\mathbb{P}(X_{t_1} \in A_{t_1}, \dots, X_{t_n} \in A_{t_n}) = \mathbb{P}(X_{t_1} \in A_{t_1}) \cdots \mathbb{P}(X_{t_n} \in A_{t_n}).$$

Attention 2.1: Independence as a Modelling Assumption



More often than not, the independence of random variables is a model *assumption*, rather than a consequence. Instead of describing a random experiment via an explicit description of Ω and $\mathbb{P}$, we will usually model the experiment through one or more (independent) random variables

2.4 Expectation

The expectation of a (numerical) random variable is a weighted average of all the values that the random variable can take. The expectation of a *discrete* random variable X is defined

as:

$$\mathbb{E}X := \sum_x x\mathbb{P}(X = x),$$

and for a *continuous* random variable X with pdf f it is defined as:

$$\mathbb{E}X := \int_{-\infty}^{\infty} xf(x)dx.$$

Having to define the expectation in two ways is unsatisfactory. Also, there are cases wherein random variables are neither discrete nor continuous. How should we define the expectation for these random variables? Here is the formal definition:

Definition 2.10: Expectation

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and X a numerical random variable. The *expectation* of X is defined as the integral

$$\mathbb{E}X := \mathbb{P}X = \int X d\mathbb{P}.$$

Theorem 2.4: Properties of an Expectation

Let $a, b \in \mathbb{R}$ and let X, Y , and (X_n) be numerical random variables for which the expectations below are well-defined. Then, the following hold:

- (1) (Monotonicity): If $X \leq Y$, then $\mathbb{E}X \leq \mathbb{E}Y$.
- (2) (Linearity): $\mathbb{E}(aX + bY) = a\mathbb{E}X + b\mathbb{E}Y$.
- (3) (Monotone convergence): If $X_n \uparrow X$, then $\mathbb{E}X_n \uparrow \mathbb{E}X$.

Lemma 2.1: Fatou

For any sequence (X_n) of positive numerical random variables,

$$\mathbb{E} \liminf X_n \leq \liminf \mathbb{E}X_n.$$

Theorem 2.5: Dominated and Bounded Convergence

Let (X_n) be a sequence of numerical random variables for which $\lim X_n = X$ exists.

Then, the following hold:

- (1) (Dominated convergence): If, for each n , $|X_n| \leq Y$ for some random variable Y with $\mathbb{E}Y < \infty$, then $\mathbb{E}X_n \rightarrow \mathbb{E}X < \infty$.
- (2) (Bounded convergence): If $|X_n| \leq c$ for all n for some $c \in \mathbb{R}$, then $\mathbb{E}X_n \rightarrow \mathbb{E}X < \infty$.

Theorem 2.6: Expectation and Image Measure

Let X be a random variable taking values in $(E, \mathcal{E})$ with distribution μ and let h be an $\mathcal{E}$ -measurable function. Then, provided the integral exists:

$$\mathbb{E}h(X) = \int h(X) d\mathbb{P} = \int_{\mathbb{R}} \mu(dx) h(x) = \mu h.$$

Theorem 2.7: Transformation Rule

Let X be a random vector with density f_X with respect to the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}^n)$. Let $Z := g(X)$, where $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible with inverse g^{-1} and Jacobian $|\frac{\partial z}{\partial x}|$. Then, at $z = g(x)$ the random vector Z has the following density with respect to the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}^n)$:

$$f_Z(z) := \frac{f_X(x)}{|\frac{\partial z}{\partial x}|} = f_X(g^{-1}(z)) \left| \frac{\partial x}{\partial z} \right|, \quad z \in \mathbb{R}^n. \quad (2.9)$$

Proposition 2.2: Expectation of a Quadratic Form

Let A be an $n \times n$ matrix and X an n -dimensional random vector with expectation vector μ and covariance matrix Σ . The random variable $Y := X^\top A X$ has expectation $\text{tr}(\Sigma A) + \mu^\top A \mu$.

2.5 L^p Spaces

Let X be a numerical random variable on

Definition 2.11: L^p Space

For each $p \in [1, \infty]$ the space L^p is comprised of all numerical random variables X for which $\|X\|_p < \infty$.

Lemma 2.2: Jensen's Inequality

Let $h : \mathcal{X} \rightarrow \mathbb{R}$ be a convex function and let X be a random variable taking values in $\mathcal{X}$, with expectation vector $\mathbb{E}X$. Then,

$$\mathbb{E}h(X) \geq h(\mathbb{E}X).$$

Theorem 2.8: Properties of the L^p Norm

Let X and Y be numerical random variables on $(\Omega, \mathcal{H}, \mathbb{P})$. Then, the following hold:

(1) (Positivity): $\|X\|_p \geq 0$, and $\|X\|_p = 0 \iff X = 0$ almost surely.

(2) (Multiplication by a constant): $\|cX\|_p = |c|\|X\|_p$.

(3) (Minkowski's (triangle) inequality):

$$|\|X\|_p - \|Y\|_p| \leq \|X + Y\|_p \leq \|X\|_p + \|Y\|_p.$$

(4) (Hölder's inequality): For $p, q, r \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = \frac{1}{r}$,

$$\|XY\|_r \leq \|X\|_p \|Y\|_q. \quad (2.10)$$

(5) (Monotonicity): If $1 \leq p < q \leq \infty$, then $\|X\|_p \leq \|X\|_q$.

2.6 Integral Transforms

Expectations are used in many probabilistic analyses.

Definition 2.12: Moment Generating Function

The *moment generating function (MGF)* of a numerical random variable X with distribution μ is the function $M : \mathbb{R} \rightarrow [0, \infty]$, given by

$$M(s) := \mathbb{E}e^{sX} = \int_{-\infty}^{\infty} \mu(dx)e^{sx}, \quad s \in \mathbb{R}.$$

Theorem 2.9: Uniqueness of the MGF

A measure μ is determined by its MGF M if the latter is finite in a neighborhood of 0.

Theorem 2.10: Taylor's Theorem for MGFs

Suppose $M(s) := \mathbb{E}e^{sX}$ is finite in the neighborhood $(-s_0, s_0)$ of 0. Then, M has the Taylor expansion

$$M(s) = \sum_{k=0}^{\infty} \frac{s^k}{k!} \mathbb{E}X^k, \quad |s| < s_0.$$

In particular, all moments $\mathbb{E}X^k, k \in \mathbb{N}$ are finite.

Theorem 2.11: Product of MGFs

If X and Y are independent, then $M_{X+Y}(s) = M_X(s)M_Y(s)$.

Definition 2.13: Characteristic Function

The *characteristic function* of a numerical random variable X with distribution μ is the function $\psi : \mathbb{R} \rightarrow \mathbb{C}$, defined by

$$\psi(r) := \mathbb{E}e^{irX} = \mathbb{E} \cos(rX) + i\mathbb{E} \sin(rX) = \int_{-\infty}^{\infty} \mu(dx)e^{irx}, \quad r \in \mathbb{R}.$$

Theorem 2.12: Inversion and Uniqueness of the Characteristic Function

If the probability measure μ has characteristic function ψ , and if $\mu\{a\} = \mu\{b\} = 0$, then

$$\mu(a, b] = \lim_{t \rightarrow \infty} \frac{1}{2\pi} \int_{\mathbb{R}} dr \mathbb{1}_{[-t, t]}(r) \frac{e^{-ira} - e^{-irb}}{ir} \psi(r) := \lim_{t \rightarrow \infty} I_t. \quad (2.11)$$

In particular, the characteristic function determines μ uniquely. If, in addition, $\int_{\mathbb{R}} dr |\psi(r)| < \infty$, then μ has a density f with respect to the Lebesgue measure, given by

$$f(x) := \frac{1}{2\pi} \int_{-\infty}^{\infty} dr e^{-irx} \psi(r). \quad (2.12)$$

Proposition 2.3: Properties of the Characteristic Function

- (1) (Symmetry): The random variables X and $-X$ are identically distributed if and only if the characteristic function of X is real-valued.
- (2) (Convolution): If X and Y are independent, then $\psi_{X+Y} = \psi_X \psi_Y$.
- (3) (Affine transformation): If a and b are real numbers, then

$$\psi_{aX+b}(r) = e^{irb} \psi_X(ar), \quad r \in \mathbb{R}.$$

- (4) (Taylor's theorem): If $\mathbb{E}|X^n| < \infty$, then

$$\psi(r) = \sum_{k=0}^n \frac{\mathbb{E}X^k}{k!} (ir)^k + O(r^n), \quad r \in \mathbb{R}.$$

2.7 Information and Independence

In this section, we have another look at the role of σ -algebras and independence in random experiments.

Definition 2.14: σ -Algebra Generated by a Random Variable

Let X be a random variable taking values in $(E, \mathcal{E})$. The sets $\{X \in A\}, A \in \mathcal{E}$ form a σ -algebra, called the σ -algebra generated by X , and we write σX .

Theorem 2.13: σ -Algebra Generated by a Stochastic Process

For the stochastic process $X := \{X_t, t \in \mathbb{T}\}$, the σ -algebra generated by X is the smallest σ -algebra on Ω that is generated by the union of the σ -algebras $\sigma X_t, t \in \mathbb{T}$. We write $\bigvee_{t \in \mathbb{T}} \sigma X_t$ or $\sigma\{X_t, t \in \mathbb{T}\}$.

Theorem 2.14: σX -Numerical Random Variables

Let X be a random variable taking values in a measurable space $(E, \mathcal{E})$. A mapping $V : \Omega \rightarrow \overline{\mathbb{R}}$ belongs to σX if and only if $V = h \circ X$ for some $h \in \mathcal{E}$.

Definition 2.15: Filtration

Let $\mathbb{T}$ be a subset of $\mathbb{R}$. A *filtration* is an increasing collection $\{\mathcal{F}_t : t \in \mathbb{T}\}$ of sub- σ -algebras of $\mathcal{H}$; that is, $\mathcal{F}_s \subseteq \mathcal{F}_t$ whenever $s < t$.

Theorem 2.15: Stochastic Process with Arbitrary Index Set

For each t in an arbitrary set $\mathbb{T}$, let X_t be a random variable taking values in a measurable set $(E_t, \mathcal{E}_t)$. Then, the mapping $V : \Omega \rightarrow \overline{\mathbb{R}}$ belongs to $\sigma\{X_t, t \in \mathbb{T}\}$ if and only if there exists a sequence (t_n) in $\mathbb{T}$ and a function h in $\bigotimes_n \mathcal{E}_{t_n}$ such that

$$V = h(X_{t_1}, X_{t_2}, \dots).$$

Definition 2.16: Independence for σ -Algebras

Let $\{\mathcal{F}_i, i = 1, \dots, n\}$ be a collection of sub- σ -algebras of $\mathcal{H}$. The $\{\mathcal{F}_i\}$ are said to be (mutually) independent or form an *independency* if

$$\mathbb{E}V_1 \cdots V_n = \mathbb{E}V_1 \cdots \mathbb{E}V_n \quad (2.13)$$

for all positive random variables $V_i \in \mathcal{F}_i, i = 1, \dots, n$.

Theorem 2.16: Independence of σ -Algebras

Sub- σ -algebras $\mathcal{F}_1, \dots, \mathcal{F}_n$ of $\mathcal{H}$ generated by p -systems $\mathcal{C}_1, \dots, \mathcal{C}_n$ are independent if and only if for all $H_i \in \mathcal{C}_i \cup \{\Omega\}$, $i = 1, \dots, n$, it holds that:

$$\mathbb{P}(H_1 \cap \dots \cap H_n) = \mathbb{P}(H_1) \dots \mathbb{P}(H_n). \quad (2.14)$$

Theorem 2.17: Independence of Random Variables

Random variables X and Y taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$ are independent if and only if

$$\mathbb{E}g(X)h(Y) = \mathbb{E}g(X)\mathbb{E}h(Y), \quad g \in \mathcal{E}_+, h \in \mathcal{F}_+. \quad (2.15)$$

2.8 Important Stochastic Processes

We conclude by highlighting some key classes of stochastic processes, including Gaussian processes, Poisson random measures, and Lévy processes.

Definition 2.17: Gaussian Process

A real-valued stochastic process $\{X_t, t \in \mathbb{T}\}$ is said to be *Gaussian* if all its finite-dimensional distributions are Gaussian (normal); that is, if the vector $[X_{t_1}, \dots, X_{t_n}]^\top$ is multivariate normal for any choice of n and $t_1, \dots, t_n \in \mathbb{T}$.

Definition 2.18: Poisson Random Measure

A random measure N on $(E, \mathcal{E})$ is said to be a *Poisson random measure* with mean measure μ if the following properties hold:

- (1) $N(A) \sim \text{Poi}(\mu(A))$ for any set $A \in \mathcal{E}$.
- (2) For any selection of disjoint sets $A_1, \dots, A_n \in \mathcal{E}$, the random variables $N(A_1), \dots, N(A_n)$ are independent.

Proposition 2.4: Poisson Random Measure on a Product Space

Let $X := \{X_i, i \in I\}$ be the atoms of a Poisson random measure N on $(E, \mathcal{E})$ with mean measure μ , and let $\{Y_i, i \in \mathbb{N}\}$ be i.i.d. random variables on $(F, \mathcal{F})$ with distribution π , independent of X . Then, $\{(X_i, Y_i), i \in I\}$ form the atoms of a Poisson random measure M on $(E \times F, \mathcal{E} \otimes \mathcal{F})$ with mean measure $\mu \otimes \pi$.

Theorem 2.18: Conditioning on the Total Number of Points

Let N be a Poisson random measure on $(E, \mathcal{E})$ with mean measure μ satisfying $\mu(E) := c < \infty$. Then, conditional on $N(E) = k$, the k atoms of N are i.i.d. with probability distribution μ/c .

Definition 2.19: Compound Poisson Process

The process $(X_t, t \geq 0)$ defined by

$$X_t := \int_{[0,t] \times \mathbb{R}^d} N(ds, dx)x, \quad t \geq 0,$$

is the *compound Poisson process* corresponding to the measure ν .

Definition 2.20: Lévy Process

The process X is said to be a *Lévy process* if

- (1) the paths of X are almost surely right-continuous and left-limited, with $X_0 = 0$,
and
- (2) for every $t_1 < t_2 < t_3 < t_4$, the increments $X_{t_2} - X_{t_1}$ and $X_{t_4} - X_{t_3}$ are independent,
and for every $t, u \in \mathbb{R}_+$, the increment $X_{t+u} - X_t$ has the same distribution as X_u .

Theorem 2.19: Lévy – It Decomposition

The process $X := (X_t, t \geq 0)$ is a Lévy process if and only if there exists a d -dimensional Brownian motion process $B := (B_t, t \geq 0)$ starting at $\mathbf{0}$ and a Poisson random measure N on $\mathbb{R}_+ \times \mathbb{R}^d$ that is independent of B and has mean measure $\text{Leb} \otimes \nu$, with ν satisfying

$$\int_{\mathbb{R}^d} \nu(d\mathbf{x}) (|\mathbf{x}|^2 \wedge 1) < \infty, \quad (2.16)$$

such that $X = B + U + Z$, where the summands are independent, $U := (U_t, t \geq 0)$ is the compound Poisson process

$$U_t := \int_0^t \int_{\|\mathbf{x}\| > 1} N(ds, d\mathbf{x}) \mathbf{x}, \quad t \geq 0,$$

and $Z := (Z_t, t \geq 0)$ is the limit of compensated compound Poisson processes:

$$Z := \lim_{\delta \downarrow 0} Z^\delta \quad \text{with} \quad Z_t^\delta := \int_0^t \int_{\delta \leq \|\mathbf{x}\| \leq 1} [\mathbf{x} N(ds, d\mathbf{x}) - ds \mathbf{x} \nu(d\mathbf{x})], \quad t \geq 0.$$

3 Convergence

4 Conditioning

5 Martingales

6 Wiener and Brownian Motion Processes

7 Itô Calculus

Part II

Levy Processes

Part III

Stochastic Volatility

Part IV

Risk-Neutral Valuation

8 Assets, Portfolios and Arbitrage

9 Discrete-time Market Model

10 Continuous-time Market Model

Part V

Martingale Methods

11 The Black-Scholes Model

12 A Dividend-paying Stock

13 Robustness of the Black-Scholes Approach

13.1 Uncertain Volatility

14 Discretisation Schemes

15 Connection to Partial Differential Equations

Part VI

Monte Carlo Methods

SIMULATION techniques are invaluable in managing complex financial scenarios that involve path-dependent payoffs or distributions without analytical expressions. One prominent application is pricing derivative securities, estimating value at risk, and emulating hedging strategies. Monte Carlo simulation, in particular, proves highly effective for such tasks, offering a blend of simplicity and adaptability as its primary advantages. However, the method's drawback lies in its considerable computational demands, which can be mitigated to some extent. Our approach employs Monte Carlo simulation to generate

simulated option price values, which are then compared against outcomes derived from analytical formulas.

16 Generating Random Numbers and Random Variables

17 Quasi Monte Carlo

18 Variance Reduction

Part VII

Grid & Lattice Methods

19 Lattice Methods

20 Finite Difference Methods

21 Contract Sensitivities

21.1 Likelihood Ratio

21.2 Pathwise Derivatives

21.3 Finite-Difference Approximations

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Financial Contracts

22 European Options

23 Exotic Options

24 American Options

25 Basket Options

Appendices

A Existence of Lebesgue Measure

B Linear Algebra and Functional Analysis

C Multivariate Differentiation

D Probability and Statistics

E Python Primer